D129 | Repetition is not rivalry evidence

PRIVATE FICTIONAL two-player symmetric stage game. Utility units, not actual company money. No competitor contact, price/subsidy coordination, business signal or financial action.

Stage game	Own payoff: mutual C=3, unilateral D against C=5, C against D=0, mutual D=1. D strictly dominates C; mutual D is unique stage Nash, while both prefer mutual C. Higher joint payoff does not identify equilibrium.
Infinite perfect monitoring	Common delta .6, fixed game, infinite horizon, both observe every past action. Grim: cooperate until any D, then mutual D forever. Cooperation3/(1-delta)=7.5; one deviation5+delta/(1-delta)=6.5. Margin1. Weak threshold .5, not unique predicted play.
Finite horizon	Known10 periods, unique stage Nash, complete information and no terminal rewards. Backward induction gives mutual D every period. Present value is ten terms starting now, not an infinite sum. Different terminal games/reputation assumptions can change this conclusion.
Noise is separate	A single public false-alarm event each still-cooperative period has independent probability.02. At least one in10=1-.98^10. The noisy policy value assumes alarm after current cooperative payoff and permanent punishment NEXT period. It is policy evaluation only, not an equilibrium test.
Signal incentives	Acceptance benefit10, strong cost4/weak14, outside0. Conditional sending gains6 and -4. Receiver payoffs, beliefs, prior and off-path behavior are not supplied; different sending incentives are not a complete signaling equilibrium.
Boundary	Actual firm payoffs, investor causal effects, exit motives and learner records unknown. Stanford written model differs in payoff table; MIT monitoring definitions selected sections only. Optional Trust landing page/repo not an executed tournament; no audio/video/full diagrams or cross-viewer/learner/account/overall10 claim.

Inputs | Utility, periods and assumptions

Synthetic symmetric payoffs. Discount is a per-period factor, not an annual interest rate. False alarm is ONE public event per cooperative period, not one independent event per player.

Temptation T: D against C, utility	5.00
Reward R: C against C, utility	3.00
Punishment P: D against D, utility	1.00
Sucker S: C against D, utility	0.00
Common per-period discount delta	0.60
Public false-alarm probability p	0.02
Known finite periods / alarm window N	10.00
Conditional acceptance benefit, utility	10.00
Strong-type signal cost	4.00
Weak-type signal cost	14.00
No-signal outside payoff	0.00

Payoff entries valid numbers?	1.000000
Discount valid, 0 <= delta < 1?	1.000000
Noise/window valid?	1.000000
Signal entries valid?	1.000000

Negative stage utility is allowed; signal benefit/cost nonnegative in this authored toy. N is1..1000 to bound the test domain, not an economic theorem. All actual model assumptions need evidence; no observed market payoffs are supplied.

Stage | Best responses before repetition

Symmetric stage game. D against C gives T; D against D gives P. Strict PD order T>R>P>S is a scope gate, not a claim about real markets.

Gain from D rather than C against C	2.000000
Gain from D rather than C against D	1.000000
Strict prisoner-dilemma order? 1=yes	1.000000

D strict best response against either? 1=yes	1.000000
Each-player gain: mutual C versus mutual D	2.000000

Baseline differences2,1; strictPD1 and dominantD1; mutualC raises each payoff2. If payoff order changes, generic arithmetic remains visible but the PD-specific grim/finite conclusions are held. MutualD Nash is not Pareto efficiency.

Infinite | Perfect-observation grim test

Fixed symmetric game, infinite horizon, common delta, perfect public observation. Unnormalized discounted sums starting with current period. MutualD is credible because D best responds to D.

Always-C path present value	7.500000
One deviation now then perpetual mutual D	6.500000
Cooperation minus deviation value	1.000000

Weak incentive threshold, strict PD only	0.500000
Normalized cooperation value	3.000000
Normalized one-deviation value	2.600000

Conditional model verdict	WEAK IC HOLDS

Baseline7.5 vs6.5, margin1, threshold.5, normalized3 vs2.6. Delta.4 fails; .5 is weak equality, not strict preference. MutualD remains another equilibrium. This does not prove real collusion/cooperation, select a unique equilibrium or establish noisy-monitoring IC.

Finite | Known final period

Only under complete information, fixed stage game with unique mutualD stage Nash, no terminal reward and known N. AllD by backward induction; stage-order gate prevents silently using it in another game.

Discounted N-period annuity starting now	2.484883
AllD finite value under scoped model	2.484883
AllC arithmetic comparator, NOT equilibrium	7.454650
Last-period gain from D against C	2.000000
Backward-induction scope status	KNOWN N: ALL D

Ten-period allD value approximates2.484883456 atdelta.6; allC comparator7.454650368. Cooperative comparator is not predicted finite equilibrium. Unknown horizon, incomplete information or multiple terminal equilibria require another model; do not generalize allD to all repeated interaction.

Noise | False alarms are not deviations

Separate policy model. One public false alarm with probability p per still-cooperative period, independent over time. Payoff R occurs before alarm, then punishment P begins next period forever. No true-deviation signal law supplied.

No false alarm in N independent opportunities	0.817073
At least one false alarm in N opportunities	0.182927

Perpetual punishment-state value	2.500000
Cooperative-state POLICY value, not IC	7.354369
Bellman identity residual	0.000000
Loss versus perfect always-C path	0.145631

Equilibrium/forgiveness verdict	POLICY ONLY: IC UNKNOWN

At least one false alarm1-.98^10=18.292719311%; not20% except a first-order approximation. Expected policy value is not an optimal policy or equilibrium. Forgiveness may reduce false punishment but also weaken deterrence; need deviation detection, public/private signals, beliefs and full strategy tests. No optimal forgiveness claim.

Signal | Conditional sender incentives only

Given acceptance follows sending and no-signal gives outside payoff. No receiver payoff, prior, belief consistency or off-path response supplied. Signal payoff is benefit minus type-specific cost.

Strong net sending payoff	6.000000
Weak net sending payoff	-4.000000
Strong gain relative to outside	6.000000
Weak gain relative to outside	-4.000000
Strict differential sending incentives?	STRONG ONLY: CONDITIONAL

Baseline strong6,weak-4,outside0. Equality is indifference, not strict separation. A higher cost is not automatically credible; both types may send or neither. Complete equilibrium requires receiver optimality and Bayes-consistent beliefs. No real business signal or competition advice.

Evidence | Match the model, not the label

Selected written sources refreshed13November2026. Model sources are not evidence of firm payoffs, conduct, investor influence or learner transfer.

Stanford, May2006	https://web.stanford.edu/~jdlevin/Econ%20286/Repeated%20Games%20I.pdf - full extracted written text read. Its stage matrix uses1,2,0,-1 and normalized sums, unlike our3,5,1,0. Both yield.5 threshold, but do not substitute its payoff values. Its finite-game caveat depends on unique stage Nash.
MIT public monitoring, Spring2024	https://ocw.mit.edu/courses/14.126-game-theory-spring-2024/mit14_126_s24_lecture_5-6_repeated_games.pdf - selected introduction and noisy-monitoring sections, not full proof/diagrams. Perfect/public/private monitoring differ; public noise can trigger costly on-path punishment. Our policy value supplies no deviation signal law.
MIT costly signaling, Spring2015	https://ocw.mit.edu/courses/15-025-game-theory-for-strategic-advantage-spring-2015/c7de229ffadd7fb5dfe1d01b52f8ea48_MIT15_025S15_Lec_19.pdf - extracted written text read. Its checklist requires receiver optimality, sender best response and consistent beliefs. Its beer/quiche and poker payoffs are not this toy.
MIT finite-game directory	https://ocw.mit.edu/courses/14-12-economic-applications-of-game-theory-fall-2012/resources/mit14_12f12_slides12/ - directory identifies finite repeated games; underlying slides not read. Not independent proof of our numbers.
Optional Trust	https://ncase.me/trust/ landing returned loading/translation navigation, not playable simulation evidence. https://github.com/ncase/trust identifies the interactive guide and source project. No tournament run, strategy ranking or audio/video coverage claimed.
Actual rivalry unknown	Ola/Uber payoffs, investor causal effects and exit motives remain unknown here. Financing,demand,regulation and firm decisions need dated primary records and a causal comparison. Stakes, price changes, exits or costly conduct alone do not identify the proposed motive. No contact,coordination,pricing or subsidy action.

Actual work | No real rivalry or learner inference

Only record performed permission-cleared work. Unknown is not zero or confirmed mastery.

Date/version	Independent stage answer	Discount/horizon objection	Noise assumption/correction	Signal missing belief	Reviewer finding	Delayed transfer

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Baseline independent formulas	PASS: all baseline formulas independently recomputed.
Changed payoff/discount scenarios	PASS:24changedsets plus restoration,including nonPD gates.
Finite horizon and weak boundary	PASS:delta .4/.5/.6 and finite N1/10/20/100.
False alarms/policy not equilibrium	PASS:independent transition-series matches noisy policy value.
Signal conditional/integer gates	PASS:signal equality,neither/both incentive scenarios.
Invalid controls and isolation	PASS:16invalidcases with unaffected model isolation.
Broken-reference holds/errors	PASS:6brokenrefs,held or explicit calculation error.
Restored baseline and actual blank	PASS:baseline restored;actual learner/reviewer/transfer blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.